

UACP: Uncertainty-Aware Channel Prediction in MIMO Networks

Abstract

The key issue with existing compression strategies for multiple-input multiple-output (MIMO) channel state information (CSI) feedback is that they rely on strong assumptions about channel statistics. Ultimately, this may lead to suboptimal CSI compression and, in turn, inaccurate MIMO precoding in case of model uncertainty and/or model misspecification. In this paper, we propose UACP, the first *uncertainty-aware* CSI prediction and compression strategy for MIMO systems. UACP is rooted in imprecise probability theory and aims at dynamically selecting the lowest number of subcarriers for channel reporting to guarantee a low CSI prediction error at the transmitter for accurate precoding. UACP uses evidential regression to obtain an estimate of the CSI prediction error in addition to the prediction itself. The key intuition is that such an error is linked with epistemic and aleatoric uncertainty of the prediction process. These two components are effectively disentangled by UACP and used as a proxy to decide the compression level, i.e., the number of subcarriers to be fed back. The results show that UACP reduces feedback overhead by 92.7% and improves throughput by 21.3% relative to Wi-Fi ($N_g=4$), while incurring only a 0.33% increase in error vector magnitude (EVM). UACP falls back to full feedback and improves bit error rate (BER) by 6.3% under distribution shift.

1 Introduction

Estimating the CSI is essential to properly design MIMO precoding and thus maximize the reception quality [7]. To do so, the majority of modern wireless networks execute an *explicit sounding* procedure before every data transmission round [13]. The channel is estimated at the receiver, and to match the direction (downlink or uplink) of the subsequent transmission, it is fed back to the transmitter for precoding. Although this approach does not require transmitter–receiver calibration, it introduces excessive feedback overhead as the CSI size grows with the number of subcarriers (determined by bandwidth) and the number of antennas [16].

To address this scalability issue, the CSI feedback is usually compressed prior to transmission. Some approaches apply learning-based data reduction on each CSI instance [2, 18, 24, 25, 27, 28], while others refrain from reporting the CSI when the channel remains static [3, 11, 17, 19, 26]. Wi-Fi networks apply uniform sampling to the subcarrier indices before feeding them back to the transmitter [6]. This technique, known as *subcarrier grouping*, assumes that CSI in adjacent subcarriers is highly correlated, so the transmitter *predicts* the CSI on the unreported subcarriers through interpolation. The authors in [22, 27] also propose to integrate this uniform subcarrier selection with learning-based data reduction. However, these approaches adopt fixed sampling rates and thus do not account for the dynamic nature of communication environments that affects the coherence bandwidth of channels.

A common shortcoming of these approaches is that they rely on deterministic or even fixed channel statistics. In this paper, we address this limitation with *Uncertainty-Aware Channel Prediction* (UACP), overviewed in Figure 1, which (1) dynamically selects the subcarriers to feed back and (2) predicts the full CSI from the sparse

representation reported. The key idea in UACP is that all channel models are inherently imperfect and unable to fully represent the actual channels. UACP captures this by using imprecise probability theory [21] and disentangles *epistemic uncertainty* from *aleatoric uncertainty* in the CSI estimation process. The former is associated with the model and determines whether the learned model can deal with the ongoing channel distribution or not. The latter is instead associated with the data and increases when the channel data becomes harder to predict. By separating these two contributions, UACP dynamically adapts to channel variations, thus reducing feedback overhead when aleatoric uncertainty decreases, and adapts the model when epistemic uncertainty increases.

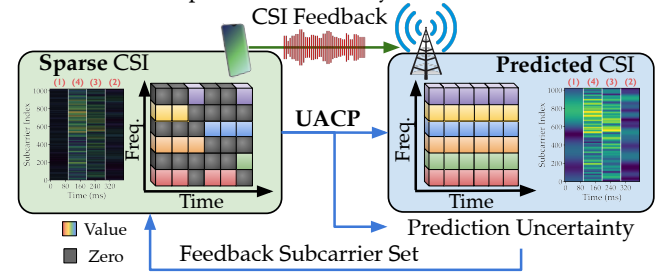


Figure 1: At every sounding round, UACP identifies the set of subcarriers to feed back for accurate CSI prediction based on prediction uncertainty. The red labels indicate the level of frequency selectivity of the channel.

Recently, a non-uniform dynamic subcarrier selection approach, called D-Anchor in this paper, has been proposed [4]. The receiver first estimates the *full* CSI and then selects the subcarriers to be fed back to minimize the *expected* CSI interpolation error at the transmitter. Instead, UACP only requires the receiver to estimate the CSI on a *subset* of subcarriers requested by the transmitter. These are selected based on the *actual* prediction uncertainty, which is obtained by the transmitter together with CSI prediction. Instead of relying on simple interpolation, UACP uses an evidential regressor that captures the underlying statistics of signal propagation and enables more accurate CSI reconstruction even with higher levels of compression than existing deterministic approaches such as D-Anchor or the uniform subcarrier subsampling performed by Wi-Fi. This is shown in the example results in Figure 2. Wi-Fi, D-Anchor, and UACP report 64, 16, and 4 out of 256 subcarriers, respectively, in the low frequency-selectivity scenario (top), while report 64, 16, and 16 subcarriers, respectively, in high frequency-selectivity (bottom). UACP achieves 52% and 23% reduction in normalized mean squared error (NMSE) compared to D-Anchor and Wi-Fi uniform subsampling, respectively.

Summary of Novel Contributions

- We devise UACP, the first channel prediction algorithm that can quantify uncertainty about its prediction, and decide on the parameters for the next sounding round accordingly.
- We account for the dynamic nature of wireless channels by assuming that the channel statistics are random themselves, thus assuming a distribution over them. This, in turn, allows for the disentanglement of the aleatoric and epistemic uncertainty.

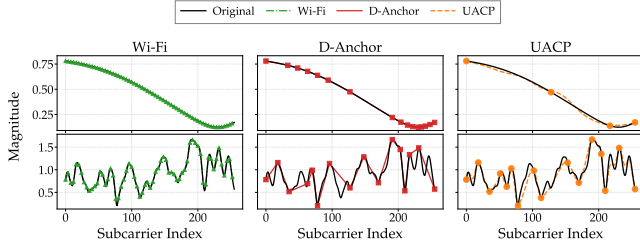


Figure 2: Ground-truth CSI and predictions using the base-lines (Wi-Fi [6] and D-Anchor [4]) and UACP in frequency-flat (top) and frequency-selective (bottom) scenarios.

- We extensively evaluated UACP using Sionna [8]. Our results show that UACP reduces airtime overhead by 92.7% and increases effective throughput by 21.3% over Wi-Fi ($N_g=4$), at only a 0.33% average EVM cost. When the channel shifts out of distribution, UACP safely reverts to full feedback and slightly lowers BER by 6.3% relative to the full-feedback oracle.

Summary of Impact

UACP enables principled selection of the feedback subcarrier set that guarantees high communication performance while minimizing feedback overhead. By addressing the key bottleneck of MIMO, UACP unlocks the use of multiple antennas to increase the number of simultaneously transmitted streams, and, in turn, the network throughput.

2 Background and Motivation

A key issue with traditional channel prediction techniques is that they typically output a single estimate of the CSI [7]. In the sparse-feedback setting considered here, this is particularly problematic. Indeed, when the transmitter infers the full CSI from a small subset of reported subcarriers, two different situations can produce identical point-estimate prediction errors while demanding fundamentally different corrective actions. The first case is when the channel is well-represented by the training distribution, but the data is intrinsically harder to predict (aleatoric uncertainty). To reduce this uncertainty in prediction, the transmitter should ask the receiver to feed back more subcarriers. The second case is when the propagation regime itself has drifted away from what the predictor has learned (epistemic uncertainty). In this case, the transmitter can no longer rely on the current model and should initiate an adaptation phase. An effective adaptive feedback mechanism must distinguish these two cases. Yet, existing channel models that only provide point estimates cannot do so. More concretely, we need to encode whether the channel predictor is uncertain due to a harder-to-predict data distribution (aleatoric uncertainty) or a substantial lack of knowledge in the prediction model (epistemic uncertainty).

However, disentangling these two uncertainty contributions is fundamentally impossible with existing channel models. Indeed, modeling approaches implicitly assume that channel statistics are fixed [7]. For example, in a classical additive white Gaussian noise (AWGN) channel model, the noise is random, having a Gaussian distribution with zero mean and fixed, yet unknown, variance (noise power). Similarly, in simplified rich-scattering MIMO orthogonal

frequency-division multiplexing (OFDM) models, the channel frequency response (CFR) per each antenna pair is often modeled as a complex Gaussian vector with zero mean in Rayleigh fading (no direct path), or non-zero mean in Rician fading (line of sight), and a fixed covariance matrix. This approach projects all uncertainty to aleatoric, while leaving the model uncertainty unrepresented. To quantify epistemic uncertainty, in UACP, we consider that even the parameters of these Gaussian distributions are stochastic themselves, randomized according to their own distribution, called the *evidential distribution*. The goal of our channel predictor is therefore to find such distribution parameters.

To do so, UACP channel modeling is based on the evidential regression theory [12, 23]. We use a deep neural network (DNN) (described in detail in Section 3.2) to infer the parameters of the evidential distributions over the parameters of the aforementioned Gaussian distributions. Unlike point-estimate methods, UACP – by leveraging these higher-order evidential distributions – yields disentangled measures of aleatoric and epistemic uncertainty in closed form. As detailed in Section 3, these quantities allow UACP to adapt the feedback subcarrier set to ongoing channel conditions and to adapt the model when it is necessary.

3 UACP Framework

We consider a general MIMO OFDM wireless communication system with N_t transmit antennas, N_r receive antennas, and K subcarriers. Let $\mathcal{K} \triangleq \{1, \dots, K\}$ denote the subcarrier index set, and let $\mathbf{H}_u \in \mathbb{C}^{K \times N_r \times N_t}$ denote the downlink CFR matrix during sounding interval u , with $k \in \mathcal{K}$. As shown in Figure 3, the transmitter initiates explicit sounding by sending a sounding packet containing pilot symbols which are known to the receiver (Step 1 in the figure). By comparing the received pilots with the known ones, the receiver obtains an estimate of the CFR. In UACP, the estimate is obtained only for a subset of subcarriers $\mathcal{S}_u \subseteq \mathcal{K}$ (Step 2), denoted by $\tilde{\mathbf{H}}_u \in \mathbb{C}^{|\mathcal{S}_u| \times N_r \times N_t}$, to reduce the feedback overhead as this estimate has to be fed back to the transmitter (Step 3). As will be discussed in Section 6, current approaches for feedback subcarrier selection use fixed subcarrier sets [15], or are based on heuristic strategies [4]. To make matters worse, *these approaches do not consider the end-to-end communication performance reached after computing the precoding weights using their approximations, which is challenging as performance metrics can only be measured after actual data transmission.*

UACP addresses this challenge by using the prediction uncertainty as a principled proxy for wireless system-level performance. Specifically, while predicting the full CFR $\hat{\mathbf{H}}_u \in \mathbb{C}^{K \times N_r \times N_t}$, the transmitter also estimates its predictive uncertainty parameters Θ_u (Step 4), which are later used to find a surrogate measure for system-level performance. These parameters are also used to optimize how many ($|\mathcal{S}_{u+1}|$) and which ($\mathcal{S}_{u+1}$) subcarriers the receiver should report in the next sounding round. In this way, UACP uses the predictive uncertainty at round u to decide on the subcarriers for the next round $u+1$. This is reasonable because sounding is performed frequently in wireless networks to keep the information available to the transmitter for precoding updated. For example, in Wi-Fi networks, sounding is performed every 10 ms [6]. The transmitter uses the predicted CFR to precode the subsequent data

packets (Step 5). Traditional postcoding (equalization) is applied at the receiver to retrieve the transmitted data (Step 6).

3.1 Main Idea

The optimal subcarrier set to be reported for channel prediction must balance two competing terms: feedback efficiency and communication performance. On the one hand, reducing $|\mathcal{S}|$ lowers the feedback overhead. On the other hand, reporting fewer subcarriers makes the transmitter rely more heavily on learning-based prediction, which can degrade channel-reconstruction accuracy and, in turn, precoding performance. This trade-off is captured by

$$S_{u+1} = \arg \min_{S \subseteq \mathcal{K}} -\mathbb{E}\left[R(\hat{\mathbf{H}}_u)\right] + \lambda_{fb}|S|, \quad (1)$$

where $\hat{\mathbf{H}}_u$ denotes the full CFR predicted at the transmitter using $\tilde{\mathbf{H}}_u$, $R(\cdot)$ represents a performance metric to be maximized, and $\lambda_{fb} > 0$ weights the feedback size term to enforce sparsity.

Ideally, $R(\cdot)$ is either a reconstruction quality metric or a communication performance metric. However, as discussed in Section 3, neither metric is directly available at the transmitter after deployment: For example, BER can only be measured after actual data reception, and the ground-truth channel $\mathbf{H}_u$ is never observable. As such, UACP uses *confidence metrics* as surrogate measures for these metrics. This requires developing an uncertainty-aware channel estimation strategy which allows disentangling *aleatoric* and *epistemic* uncertainty. In this way, the transmitter obtains both the complete CFR and a confidence score for the prediction quality that allows dynamic adaptation of S_u .

How UACP uses aleatoric uncertainty. Using aleatoric uncertainty, UACP can determine *how much* CSI feedback is needed. If the average aleatoric uncertainty over the dropped subcarriers is small, the adopted sparsity at sounding round u is sufficient, and the next feedback can be made more compact. If aleatoric uncertainty grows, UACP requests more subcarriers in the next round. Also, UACP can determine *which* frequency responses should be fed back. Specifically, the uncertainty map is generally non-uniform, especially in highly frequency-selective channels. The transmitter will thus select subcarriers that contribute more to the prediction risk. Overall, aleatoric uncertainty allows identifying the set of subcarriers S_u for which the CFR estimate should be requested from the receiver.

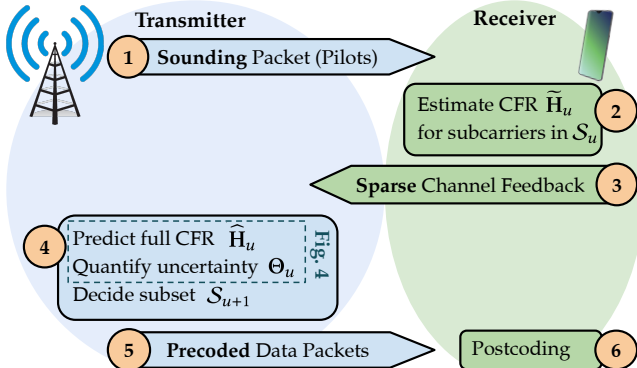


Figure 3: Overview of UACP channel compression (at the receiver) and prediction (at the transmitter).

How UACP uses epistemic uncertainty. When operating in environments similar to those used to train the predictor, the sparse observations provide enough confidence for accurate estimation. Therefore, the epistemic uncertainty remains low, which means that the trained DNN can support the current regime. In contrast, when the propagation regime changes substantially such that the new channel no longer matches the learned distribution well, the DNN reacts by lowering its confidence. Subsequently, the epistemic uncertainty grows, signaling the need for more training/adaptation. While adapting the predictor, UACP enlarges the feedback subset to the maximum ($S_u = \mathcal{K}$) to maintain adequate system-level performance.

Algorithm 1 summarizes how UACP operates. In the following, we detail the channel prediction strategy in Section 3.2, while the adaptive subcarrier selection strategy is discussed in Section 3.3.

Algorithm 1: UACP channel prediction, feedback adaptation

Input: Subcarrier set S_u ; sparse CFR observation $\mathbf{Z}_u$.

Output: Predicted full CFR $\hat{\mathbf{H}}_u$; next subcarrier set S_{u+1} .

- 1: **Step 1: Evidential Regression (Section 3.2)**
 - 2: Execute the evidential model $\Theta_u \leftarrow f_\phi(\mathbf{Z}_u, \mathbf{M}_u)$
 - 3: Obtain the predicted full CFR $\hat{\mathbf{H}}_u$ using Equation (6)
 - 4: Obtain the aleatoric and epistemic uncertainty covariance according to Equations (7)-(8)
 - 5: **Step 2: Evaluate Feedback Relevance (Section 3.3.1)**
 - 6: Compute the epistemic uncertainty score $\mathcal{U}_u^{\text{epi}}(\tilde{S}_u)$ using Equation (13)
 - 7: **if** $\mathcal{U}_u^{\text{epi}}(\tilde{S}_u) > \mathcal{U}_{\max}^{\text{epi}}$ **then**
 - 8: Request all subcarriers for the next round $S_{u+1} \leftarrow \mathcal{K}$
 - 9: Trigger model adaptation
 - 10: **return** $\hat{\mathbf{H}}_u, S_{u+1}$
 - 11: **end**
 - 12: **Step 2-a: Subcarrier Rate Selection Only (Section 3.3.2)**
 // Uniform Subsampling
 - 13: Compute the aleatoric uncertainty score $\mathcal{U}_u^{\text{ale}}(\tilde{S}_u)$ using Equation (13)
 - 14: **if** $\mathcal{U}_u^{\text{ale}}(\tilde{S}_u) < \mathcal{U}_{\text{opt}}^{\text{ale}} - \Delta$ **then**
 - 15: Decrement rate
 - 16: **else if** $\mathcal{U}_u^{\text{ale}}(\tilde{S}_u) > \mathcal{U}_{\text{opt}}^{\text{ale}} + \Delta$ **then**
 - 17: Increment rate
 - 18: **end**
 - 19: **Step 2-b: Full Subcarrier Set Selection (Section 3.3.3)**
 // Non-uniform Subsampling
 - 20: Select the next subcarrier set S_{u+1} using Equation (14)
 - 21: **return** $\hat{\mathbf{H}}_u, S_{u+1}$
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3.2 Evidential Regression for Channel Prediction

Channel prediction in UACP is formulated as an evidential regression problem [1, 14, 20]. This choice enables UACP to represent uncertainty over the channel statistics and to disentangle aleatoric

and epistemic uncertainty. Evidential regression provides both the predicted CFR and uncertainty measures, which are the key signals to adjust the feedback sparsity, in a single forward pass of the DNN. Therefore, unlike Bayesian or sampling-based uncertainty-quantification approaches [5, 10] that typically require multiple stochastic forward passes or costly posterior inference, evidential regression introduces only a modest computational overhead compared to deterministic prediction [23], making it particularly appealing in our real-time and resource-constrained setting.

Let $m_u[k] \in \{0, 1\}$ denote a binary mask indicating whether subcarrier k is reported at time u , i.e., $m_u[k] = 1$ if $k \in \mathcal{S}_u$ and $m_u[k] = 0$ otherwise. Without loss of generality, we assume the same subcarriers are dropped across different antenna pairs, and use $\mathbf{M}_u$ to denote the mask tensor replicated across antenna dimensions. A sparse channel representation is obtained at the transmitter $\mathbf{Z}_u \in \mathbb{C}^{K \times N_r \times N_t}$ by embedding $\tilde{\mathbf{H}}_u$ into the full subcarrier grid and assigning zero entries to unreported subcarriers. As shown in Figure 4, given the sparse representation $\mathbf{Z}_u$, the transmitter executes a mask-aware evidential regression DNN f_ϕ to predict the full CFR $\hat{\mathbf{H}}_u$. The architecture of f_ϕ can be designed based on a deterministic regression backbone, while adding an evidential head for uncertainty quantification. The backbone can be instantiated as a convolutional neural network (CNN), a transformer, or a hybrid model. We will refer to the UACP CFR prediction as “channel inpainting”, inspired by image inpainting in computer vision literature [9]. Note that both the sparse CFR $\mathbf{Z}_u$ and the full CFR $\hat{\mathbf{H}}_u$ are complex matrices and need to be converted to (back from) their equivalent real-valued matrices before (after) being fed to f_ϕ . This is done by concatenating the real and imaginary parts in a single real-valued matrix. To simplify formulation, in the following, we consider all operations are done in the real domain.

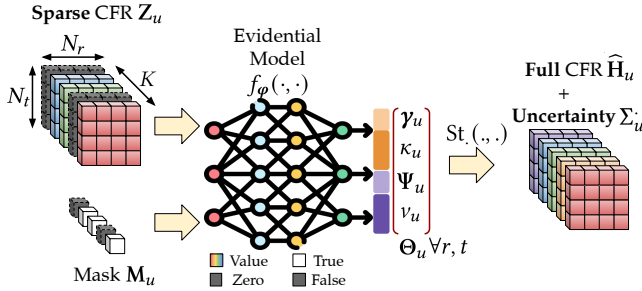


Figure 4: Evidential channel estimation model in UACP.

A fully joint multivariate evidential model over all entries of the matrix $\mathbf{H}_u$ would be prohibitively expensive, since the output dimensionality and the number of covariance parameters would scale with the multicarrier-MIMO dimensionality ($K \times N_r \times N_t$). To avoid this bottleneck while preserving frequency-domain correlation between different subcarriers, we factorize the model across receiver-transmitter antenna pairs, assigning a separate multivariate Gaussian distribution to each pair (r, t) . Let $\mathbf{h}_{u,r,t} \in \mathbb{R}^{2K}$ represent the CFR vector in the sounding round u and for receiver antenna r and transmitter antenna t over all subcarriers and converted into a real-valued vector. Let $\mathcal{P} \triangleq \{(r, t) : r = 1, \dots, N_r, t = 1, \dots, N_t\}$ denote the set of all receiver-transmitter pairs. We assume that, conditioned on $\mathbf{Z}_u$, the CFR vectors for different pairs are independent, and the conditional joint probability $p(\{\mathbf{h}_{u,r,t}\}_{(r,t) \in \mathcal{P}} | \mathbf{Z}_u)$

can be approximated as

$$p(\{\mathbf{h}_{u,r,t}\}_{(r,t) \in \mathcal{P}} | \mathbf{Z}_u) \approx \prod_{(r,t) \in \mathcal{P}} p(\mathbf{h}_{u,r,t} | \mathbf{Z}_u), \quad (2)$$

which preserves frequency-domain correlation within each antenna pair while avoiding monolithic covariance across CFR entries.

3.2.1 Evidential Modeling. We model the per-antenna-pair CFR $\mathbf{h}_{u,r,t}$ as a Gaussian random vector with mean $\boldsymbol{\mu}_{u,r,t} \in \mathbb{R}^{2K}$ and covariance matrix $\Sigma_{u,r,t} \in \mathbb{R}^{2K \times 2K}$, i.e.,

$$\mathbf{h}_{u,r,t} \sim \mathcal{N}(\boldsymbol{\mu}_{u,r,t}, \Sigma_{u,r,t}). \quad (3)$$

As discussed in Section 2, we assume even the parameters $\boldsymbol{\mu}_{u,r,t}$ and $\Sigma_{u,r,t}$ are stochastic. Specifically, the mean takes a Gaussian form, while the covariance matrix assumes an inverse-Wishart distribution [12]. This leads to a normal-inverse-Wishart (NIW) evidential distribution, which serves as a prior over these parameters, i.e.,

$$(\boldsymbol{\mu}_{u,r,t}, \Sigma_{u,r,t}) | \mathbf{Z}_u \sim \text{NIW}(\boldsymbol{\gamma}_{u,r,t}, \kappa_{u,r,t}, \Psi_{u,r,t}, \nu_{u,r,t}), \quad (4)$$

where $\boldsymbol{\gamma}_{u,r,t}$ is the prior mean (location), $\kappa_{u,r,t} > 0$ controls the confidence in the mean, $\Psi_{u,r,t} > \mathbf{0}$ is the scale matrix, and $\nu_{u,r,t} > 2K + 1$ controls the confidence in the covariance. The evidential model allows obtaining an estimate of the parameters of the distribution in Equation 4 starting from the sparse CFR observation $\mathbf{Z}_u$ and the mask $\mathbf{M}_u$. Defining $\Theta_{u,r,t} \triangleq [\boldsymbol{\gamma}_{u,r,t}, \kappa_{u,r,t}, \Psi_{u,r,t}, \nu_{u,r,t}]$, we have $\Theta_u = f_\phi(\mathbf{Z}_u, \mathbf{M}_u)$ (line 2 in Algorithm 1). Using the mask as input allows the evidential DNN to explicitly know which frequencies were observed and which ones must be inferred.

These derived parameters are used to obtain the full CFR prediction together with the prediction uncertainty. Under Equation (3) as the likelihood and Equation (4) as the prior, the posterior predictive density per CFR antenna-pair in Equation (2) is a multivariate Student- t distribution that could be written as

$$p(\mathbf{h}_{u,r,t} | \mathbf{Z}_u) = \text{St}_{\nu_{u,r,t} - 2K + 1} \left(\boldsymbol{\gamma}_{u,r,t}, \frac{\kappa_{u,r,t} + 1}{\kappa_{u,r,t} (\nu_{u,r,t} - 2K + 1)} \Psi_{u,r,t} \right). \quad (5)$$

Based on this, the predictive CFR mean, and aleatoric and epistemic uncertainty covariances are available in closed form (lines 3-4 in Algorithm 1):

$$\hat{\mathbf{h}}_{u,r,t} = \mathbb{E}[\mathbf{h}_{u,r,t} | \mathbf{Z}_u] = \boldsymbol{\gamma}_{u,r,t}, \quad (6)$$

$$\Sigma_{u,r,t}^{\text{ale}} = \mathbb{E}[\Sigma_{u,r,t} | \mathbf{Z}_u] = \frac{\Psi_{u,r,t}}{\nu_{u,r,t} - 2K - 1}, \quad (7)$$

$$\Sigma_{u,r,t}^{\text{epi}} = \text{Var}[\boldsymbol{\mu}_{u,r,t} | \mathbf{Z}_u] = \frac{1}{\kappa_{u,r,t}} \Sigma_{u,r,t}^{\text{ale}}. \quad (8)$$

Equation (7) is known to capture the uncertainty associated with the data-generating process, whereas Equation (8) captures the uncertainty due to insufficient evidence. As detailed in Section 3.1, this distinction allows the transmitter to take meaningful actions accordingly. If the aleatoric uncertainty increases, i.e., the system is transitioned to a harder-to-predict environment, the transmitter will request more subcarriers for the next feedback. On the other hand, a high value for the epistemic term reveals that the environmental change is beyond what the system has been trained for, and thus, UACP will trigger adaptation.

Note that to ensure numerical stability, the evidential DNN has to output valid values for $\Psi_{u,r,t}$, $\kappa_{u,r,t}$, and $\nu_{u,r,t}$. Specifically,

to guarantee that the matrix $\Psi_{u,r,t}$ is positive semi-definite, i.e., $\Psi_{u,r,t} \succeq \mathbf{0}$, we parameterize it through a lower-triangular matrix: $\Psi_{u,r,t} = \mathbf{L}_{u,r,t} \mathbf{L}_{u,r,t}^\top$, where the diagonal entries of $\mathbf{L}_{u,r,t}$ are exponentiated or passed through a `softplus` activation to enforce them to be positive. Similarly, positivity and admissibility of other uncertainty parameters are enforced by

$$\kappa_{u,r,t} = \text{softplus}(\tilde{\kappa}_{u,r,t}) + \epsilon, \quad v_{u,r,t} = 2K + 1 + \text{softplus}(\tilde{v}_{u,r,t}) + \epsilon, \quad (9)$$

for a small $\epsilon > 0$. The evidential DNN therefore produce $\mathbf{L}_{u,r,t}$, $\tilde{\kappa}_{u,r,t}$, and $\tilde{v}_{u,r,t}$, instead of directly providing $\Psi_{u,r,t}$, $\kappa_{u,r,t}$, and $v_{u,r,t}$.

3.2.2 Evidential Training. UACP training aims at finding the DNN parameters ϕ such that the evidential model learns to perform CFR inpainting. For each training sample consisting of a full CFR, we randomly draw a subset mask $\mathbf{M}$ from all the possible subsets of $\mathcal{K}$ so that the evidential DNN is exposed to different sparsity patterns during training. This is essential for learning a mask-conditional channel inpainting function that generalizes beyond a single subcarrier rate, rather than overfitting to a fixed feedback density.

In our framework, the training loss function is composed of two terms. First, we maximize the model evidence, which is equivalent to minimizing the negative log-likelihood (NLL) of the Student- t predictive density. Summing across sounding samples and antenna pairs yields

$$\mathcal{L}_{\text{NLL}} = -\sum_u \sum_{(r,t) \in \mathcal{P}} \log p(\mathbf{h}_{u,r,t} | \mathbf{Z}_u), \quad (10)$$

where the predictive posterior density is reported in Equation (5). Second, to discourage the evidential model from assigning strong evidence to incorrect predictions, we include an evidence-alignment regularizer in the loss function:

$$\mathcal{L}_{\text{reg}} = \sum_u \sum_{(r,t) \in \mathcal{P}} \left\| (\mathbf{h}_{u,r,t} - \mathbf{y}_{u,r,t}) \right\|_2 \cdot \Phi_{u,r,t}, \quad (11)$$

where $\Phi_{u,r,t} \triangleq \kappa_{u,r,t} + v_{u,r,t}$ is a scalar proxy for virtual evidence. Note that the regularizer is applied to all subcarriers – both the omitted and the observed frequencies. The complete loss is finally obtained as $\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$, where λ_{reg} determines the relative importance of the regularization term. Overall, the evidential model learns to perform two tasks: inpainting the unreported frequency responses and denoising/improving the explicitly fed-back subcarriers. The first term in the loss function encourages the predictive Student- t density to model the full channel matrix, while the second term suppresses evidence when the prediction error is large. Therefore, it plays an important role in preventing the evidential DNN from becoming overconfident when the sparse observations are insufficient for reliable prediction.

3.3 Adaptive Subcarrier Selection

Let $\Sigma_{u,r,t} \in \mathbb{R}^{2K \times 2K}$ denote a predictive uncertainty covariance – either aleatoric $\Sigma_{u,r,t}^{\text{ale}}$, epistemic $\Sigma_{u,r,t}^{\text{epi}}$, or total uncertainty covariance $\Sigma_{u,r,t}^{\text{tot}}$ which is the summation – for antenna pair (r, t) at interval u . For each subcarrier k , let $\mathbf{P}_k$ be the binary selection matrix (0–1 valued) that extracts the real and imaginary entries associated with subcarrier k from the real-valued vector $\mathbf{h}_{u,r,t}$. We define a

subcarrier-level uncertainty map as

$$\mathcal{U}_u[k] \triangleq \frac{1}{N_r N_t} \sum_{(r,t) \in \mathcal{P}} \text{tr}(\mathbf{P}_k \Sigma_{u,r,t} \mathbf{P}_k^\top). \quad (12)$$

Intuitively, $\mathcal{U}_u[k]$ quantifies how risky it is to leave subcarrier k unreported in the next sounding interval. As such, large values indicate frequencies whose values can no longer be confidently predicted from the currently observed sparse pattern.

We also define a normalized uncertainty score over the *unreported* subcarriers $\tilde{\mathcal{S}} \triangleq \mathcal{K} \setminus \mathcal{S}$ as

$$\mathcal{U}_u(\tilde{\mathcal{S}}) = \frac{1}{|\tilde{\mathcal{S}}|} \sum_{k \in \tilde{\mathcal{S}}} \mathcal{U}_u[k]. \quad (13)$$

3.3.1 Identifying Feedback Relevance. UACP uses the epistemic uncertainty score over the currently-unreported subcarriers, i.e., $\mathcal{U}_u^{\text{epi}}(\tilde{\mathcal{S}}_u)$ (line 6 in Algorithm 1), to decide whether the ongoing communication environment can be handled by the learned evidential model or it departed from the training distribution. If $\mathcal{U}_u^{\text{epi}}$ is high, i.e., above an environment-specific threshold $\mathcal{U}_{\text{max}}^{\text{epi}}$ (line 7 in Algorithm 1), the full CFR cannot be reliably predicted by the channel inpainting DNN due to a potentially major distribution shift in the propagation regime. Hence, UACP triggers the adaptation of the evidential model, while requesting that all subcarriers be reported in the next round, i.e., $\mathcal{S}_{u+1} = \mathcal{K}$ (lines 8-9). Instead, if the epistemic uncertainty score is low, UACP uses the full CFR prediction for precoding. In this case, UACP can be used either to select the subcarrier density only or to select both the density and placement of the subcarriers as detailed next.

3.3.2 Selecting Subcarrier Rate Only. Modern standards such as Wi-Fi (IEEE 802.11) only support uniform subsampling of subcarriers. If we want to keep UACP standard-compliant, we can limit the search to adapting the density of subcarriers only – called subcarrier grouping in Wi-Fi. Let $N_g = K/|\mathcal{S}|$ denote the subcarrier grouping factor, which is the reciprocal of the subcarrier rate. To optimize N_g , UACP evaluates the risk of dropping the same subcarriers again for the next round. Specifically, UACP computes the aleatoric uncertainty score over the currently-dropped set, i.e., $\mathcal{U}_u^{\text{ale}}(\tilde{\mathcal{S}}_u)$ (line 13 in Algorithm 1). The aleatoric uncertainty is small when the current sparse feedback is sufficient for confident inpainting, and large when the channel becomes more difficult to reconstruct, e.g., due to a reduced coherence bandwidth or stronger frequency selectivity.

Based on this, UACP updates the subcarrier rate for the next round by comparing $\mathcal{U}_u^{\text{ale}}$ against a target level $\mathcal{U}_{\text{opt}}^{\text{ale}}$ with a hysteresis margin $\Delta^{\text{ale}} > 0$. In frequency-smooth channels, the evidential DNN estimates the missing frequencies confidently, so the optimal subset for the next sounding round $\mathcal{S}_{u+1}$ remains equally sparse or can be made even more compact (lines 14-15 in Algorithm 1). Instead, when the channel becomes more frequency-selective, the aleatoric uncertainty score on the dropped frequencies grows, which immediately pushes UACP towards requesting a denser feedback (lines 16-17 in Algorithm 1). As an example, assume we have an ordered set of candidate numbers of subcarriers that can be fed back. When the uncertainty on the omitted subcarriers falls below the lower bound $\mathcal{U}_{\text{opt}}^{\text{ale}} - \Delta$, UACP moves to the next sparser candidate and reduces feedback overhead. Instead, when the uncertainty exceeds the upper bound $\mathcal{U}_{\text{opt}}^{\text{ale}} + \Delta$, UACP moves to the next

denser candidate requesting more subcarriers. Finally, if the score remains within the hysteresis interval, the current subcarrier rate is preserved for the next sounding round.

3.3.3 Selecting Full Subcarrier Set. UACP can also be used to select both the density and the placement of subcarriers. Specifically, the subcarrier set to be fed back in the next sounding round can be found by solving the following minimization problem

$$\mathcal{S}_{u+1} = \arg \min_{\mathcal{S} \subseteq \mathcal{K}} \mathcal{U}_u^{\text{ale}}(\bar{\mathcal{S}}) + \lambda_{\text{fb}} |\mathcal{S}|, \quad (14)$$

where the first term is a surrogate measure for the first term in Equation (1) and serves as a model-based proxy for the penalty of *not* feeding back a set of frequencies ($\bar{\mathcal{S}}$), while the second term is the cost associated with explicit sounding overhead. To solve Equation (14), UACP starts from all subcarriers ($\mathcal{S} = \mathcal{K}$), and evaluates the uncertainty map $\mathcal{U}_u^{\text{ale}}[k]$ using Equation (12). Then, it keeps dropping the frequencies with the least uncertainty contributions until the reduction in feedback cost no longer justifies the increase in the first term.

4 Evaluation Setup

We consider a multicarrier MIMO system ($N_r=2, N_t=2$) with $K=1024$ subcarriers. We implement the UACP evidential regressor as a mask-aware 1D residual convolutional inpainting network operating over the frequency axis. Each complex CFR is first converted to its equivalent real-valued representation; this representation is then concatenated with the binary feedback mask, allowing the model to explicitly encode which subcarriers are observed and which must be inferred. In the considered 2×2 MIMO setting, the resulting input has 9 channels. The backbone begins with a 1×1 convolution that lifts the input to 192 hidden channels, and is followed by 32 residual blocks, each composed of two 1D convolutions with kernel size 5, GELU nonlinearities, dropout ($p = 0.05$), and an identity skip connection. A final evidential head converts the shared feature map into the parameters of the evidential distribution. This head performs CFR prediction and uncertainty quantification through the uncertainty parameters. We enforce parameter admissibility through softplus-based transformations as described in Section 3.2.1.

We simulated the MIMO system using the physical layer (PHY) simulation pipeline of Sionna [8]. We set the carrier frequency to 3.5 GHz and the subcarrier spacing to 30 kHz. The system operates at 15 dB signal-to-noise ratio (SNR). We collected an extensive dataset of CFR samples varying the propagation conditions whose quality is characterized by the channel frequency-selectivity. Specifically, this is controlled through the root mean square (RMS) delay spread, which directly affects the coherence bandwidth and therefore the difficulty of predicting missing subcarriers from sparse observations. The training set contains 100,000 independently generated CFR realizations. For each training sample, the delay spread is drawn uniformly from the interval $[10, 100]$ ns. We train the model with a learning rate of 10^{-4} and a batch size of 4096. Training is performed for 150 epochs. To expose the model to different feedback sparsity patterns, the observation mask is randomized online during training. We set the regularization coefficient to $\lambda_{\text{reg}} = 10^{-3}$ when training the evidential model.

Evaluation Regimes. To disentangle uncertainty behavior, evaluation is performed on both in-distribution (ID) and out-of-distribution (OOD) regimes. Specifically, we consider three test conditions:

- **ID-Easy:** channels well-represented by the training dataset with relatively small delay spread (20 ns), for which the CFR is smoother over frequency and sparse feedback suffices;
- **ID-Hard:** channels still inside the training interval but with a larger delay spread (80 ns), for which the CFR is more frequency-selective, and reconstruction becomes intrinsically harder;
- **OOD:** channels outside the training interval, i.e., with delay spread exceeding 100 ns.

UACP is designed to react differently to the last two cases. Higher *aleatoric* uncertainty reflects increased prediction difficulty in ID-hard conditions, whereas higher *epistemic* uncertainty indicates that the current CFR is insufficiently represented by the training data. Each testing set contains 60,000 sequences of 64 temporally-correlated CFR realizations, drawn from a certain regime. For static-regime experiments, a single test set following the same distribution for all sounding intervals is used, and we evaluate the steady-state operating point reached by the controller. For the dynamic experiment, the propagation regime changes over time between ID-easy, ID-hard, and OOD, and we track the evolution of system-level performance and efficiency measures.

Benchmarks. We compare UACP against the following baselines:

- **Full Feedback (Oracle):** the receiver feeds back the CFR over all subcarriers. This serves as the upper bound for reconstruction quality and system performance after precoding, and as a lower bound for efficiency as it exhibits the highest feedback overhead;
- **Fixed Sparse Feedback & Interpolation (Wi-Fi):** the receiver reports a fixed uniformly-spaced subset of subcarriers and the transmitter reconstructs the missing ones using linear interpolation. This represents the Wi-Fi (IEEE 802.11) standard approach [6]. We consider two subcarrier grouping options: $N_g=4$ and $N_g=16$;
- **Dynamic Anchor Feedback & Interpolation (D-Anchor):** the receiver reports a fixed number of subcarriers, but the anchor subcarriers are selected adaptively across frequency to minimize interpolation distortion after quantization, resulting in a more effective subcarrier placement. The transmitter then reconstructs the missing subcarriers from these irregularly placed anchors [4]. We adopted two variants of D-Anchor, where the subcarrier grouping is either $N_g=4$ or $N_g=16$ (non-uniformly selected).

5 Experimental Results

We evaluate UACP from three perspectives. Our objective is to validate our thesis that using the uncertainty to adapt the feedback subset improves the efficiency-performance trade-off compared to the baselines. To do so, we answer three research questions:

- **Does the estimated confidence reflect system performance?** In particular, does the uncertainty track the reconstruction difficulty, and therefore, can we use it as a surrogate measure for system-level metrics after deployment, when the metrics are not available?
- **Does the uncertainty estimate work?** In particular, do aleatoric and epistemic uncertainty exhibit the expected behavior both when the channel becomes harder to predict *within* the training distribution and as it shifts *outside* the training distribution?

• **Does uncertainty-awareness translate into system efficiency gains?** Specifically, can UACP use its uncertainty to decide *how many* and *which* subcarriers to feed back to reduce feedback overhead while preserving communication performance?

Accordingly, we consider the following uncertainty-related and system-related metrics:

• **Correlation between Uncertainty & System Performance:** (i) Pearson correlation coefficient to measure the agreement between predictive uncertainty and actual reconstruction error; (ii) risk-coverage behavior, obtained by sorting samples according to uncertainty and measuring the prediction error on the retained low-uncertainty fraction.

• **Uncertainty Estimation Performance:** (i) separation between ID and OOD propagation regimes, evaluated through both differences in the epistemic uncertainty score distributions and area under the receiver operating characteristic (AUROC); (ii) regression calibration, evaluated through empirical coverage of nominal confidence intervals and calibration error.

• **Wireless System Performance & Efficiency:** (i) NMSE prediction accuracy between the predicted and ground truth CFR; (ii) performance metrics, i.e., EVM/BER after precoding, and throughput; (iii) efficiency metrics, i.e., data overhead, airtime overhead.

5.1 Uncertainty vs System Performance

In UACP, we use predictive uncertainty (first term in Equation (14)) as a surrogate metric for system performance (first term in Equation (1)), assuming that the predictive uncertainty produced by the evidential DNN is informative about the reconstruction quality of the missing subcarriers. To validate this assumption, we evaluate the relationship between these quantities by testing the evidential model under different channel conditions (ID-easy, ID-hard, OOD), using a subcarrier rate 1/16 for the input sparse CFR.

Uncertainty-Error Correlation. Figure 5 shows the reconstruction NMSE as a function of the total predictive uncertainty measured over the omitted spectrum, where the uncertainty score is obtained using Equation (13). Across all tested regimes, higher uncertainty is associated with worse reconstruction quality, which confirms the strong bond between predictive uncertainty and actual error. This relationship is more prominent in the ID-easy regime, where the reported correlation is 0.805, and also remains high in the ID-hard regime with correlation 0.740. Under OOD conditions, the correlation becomes slightly weaker yet relatively high (0.559). In other words, even under a distribution shift, uncertainty continues to preserve useful ranking information about the prediction error. Overall, these results validate our key intuition that the uncertainty score provides a meaningful proxy for system error.

Risk-Coverage Curve. A complementary perspective is provided in Figure 6. We sort the CFR samples in each of the three different datasets by their predicted total uncertainty and evaluate the average prediction NMSE using increasing fractions of the ordered datasets, starting from 10% (i.e., only the least uncertain CFR samples are considered) to 100% (i.e., the entire datasets are used). The results show that averaging over more samples from the low-uncertainty-ordered datasets leads to a monotonic increase in the error, which is correlated to an increase in the total uncertainty of the samples considered for the evaluation. This further confirms

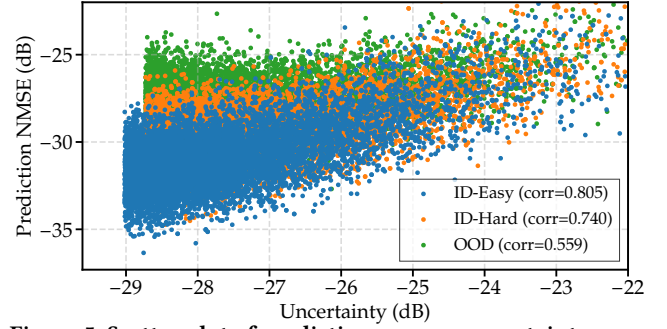


Figure 5: Scatter plot of prediction error vs uncertainty score over unreported subcarriers.

that the total uncertainty is a good proxy for the prediction NMSE and, in turn, can be used to evaluate how many and which subcarriers should be reported. Specifically, the model knows which predictions are more likely to fail, even if it cannot predict the exact magnitude of the error, and can trigger more robust feedback.

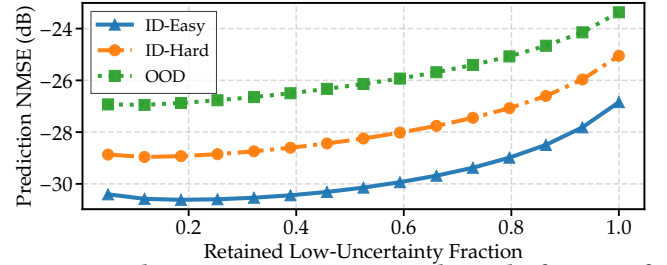


Figure 6: Risk vs coverage curve: x-axis shows the fraction of retained high-confidence samples, while y-axis is the error.

5.2 Uncertainty Estimation Performance

Next, we examine whether UACP separates the two uncertainty sources in a meaningful way, and if the uncertainty is well calibrated to be used as a control signal.

Aleatoric & Epistemic Behavior. In Figures 7 we evaluate the aleatoric and epistemic uncertainty in the ID scenario, entailing both ID-easy and ID-hard. As the delay spread increases within the training range, the average *aleatoric uncertainty* score (orange dash-dotted line in the figure) grows consistently. This reflects the increase in prediction difficulty due to the increase in the frequency-selectivity of the channel: Being the CFR less smooth in frequency, sparse prediction becomes intrinsically harder and the prediction NMSE increases (blue solid line in the figure). Instead, the epistemic component (green dotted line in Figure 7) remains relatively stable, indicating that the uncertainty is not related to model misspecification but only to the data. Indeed, the evaluation is performed in the same training regime.

A more in-depth evaluation of the epistemic uncertainty is shown in Figure 8 for the ID-easy, ID-hard and OOD datasets, where the last set is further divided into OOD-near (delay spread 120 ns) and OOD-far (delay spread 1 ms). The distribution of the epistemic uncertainty for the ID-easy and ID-hard test sets is nearly identical, further confirming the result in Figure 7. When the delay spread exceeds the training interval in the OOD cases, meaning that the propagation statistics shift drastically, the *epistemic* uncertainty produced by the evidential model increases substantially

with respect to both ID sets. This informs the transmitter that the previously learned channel structure is no longer valid and, in turn, the evidential prediction cannot be reliably used for precoding. Overall, this yields to a strong separability between ID and OOD when using epistemic uncertainty as the classification statistic with an AUROC of 0.985. As such, UACP can distinguish channels that are inherently harder to predict and thus require a denser feedback regime and therefore deserve a more conservative control action, such as further adapting the evidential model in an online mode.

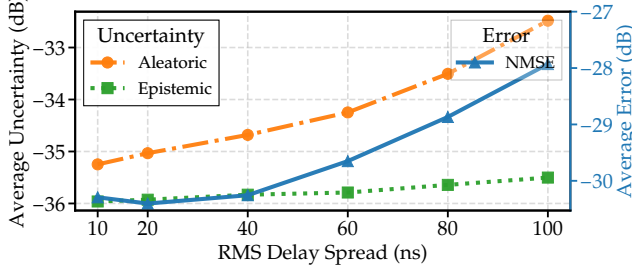


Figure 7: Average aleatoric/epistemic uncertainty and reconstruction NMSE when varying the delay spread within the training interval.

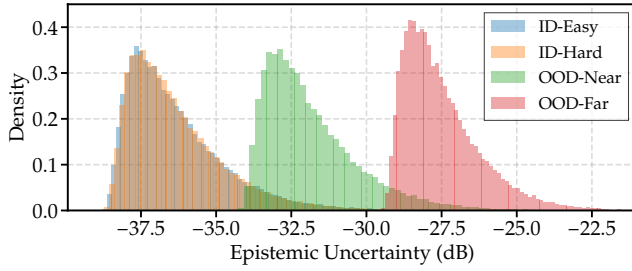


Figure 8: Distribution function of epistemic uncertainty for ID vs OOD channels (AUROC=0.985).

Calibration. Figure 9 reports regression calibration curves obtained by converting the predictive (evidential) distribution into confidence intervals and comparing the expected (nominal) and the observed (empirical) coverage. The curve remains close to the diagonal (ideal) one in ID conditions, with a calibration error of 0.103, which indicates that the confidence output of UACP is statistically significant. The calibration naturally degrades under OOD shift, with a calibration error of 0.170, but even in this case the model remains conservative in the sense that higher uncertainty still corresponds to higher risk.

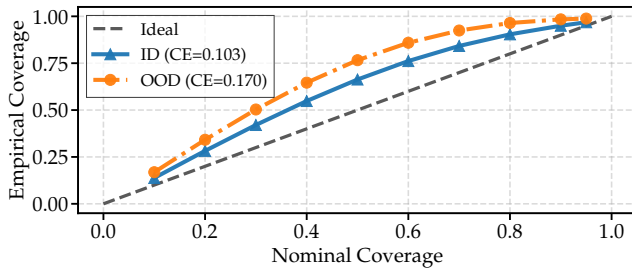


Figure 9: Interval coverage plot for ID and OOD conditions.

Taken together, the previous results establish that UACP uncertainty is: (i) semantically decomposed into in-distribution hardness and distributional mismatch; and (ii) calibrated well enough to drive feedback rate control.

5.3 Wireless System Performance

We now move from uncertainty quality to communication performance and answer the third research question about whether uncertainty-aware feedback adaptation improves the trade-off between uncertainty-aware feedback adaptation improves the trade-off between feedback overhead and downstream wireless performance.

Figures 10a and 10b compare *communication performance* achieved by UACP and the baselines in terms of BER and EVM, while Figure 10c compares *communication efficiency* in terms of feedback airtime. Also, the *prediction performance* of all methods in terms of NMSE is reported in Figure 10d. Moreover, Table 1 shows the effective throughput after accounting for the overhead reduction. Overall, the results highlights that UACP offers a better trade-off point between performance and efficiency compared to other baselines. The reason is that UACP does not commit to a single feedback density. Specifically, it uses a highly-sparse feedback when the channel is easy to predict, increases the reported subcarriers as frequency-selectivity grows, and falls back to full feedback when epistemic uncertainty indicates an OOD regime.

In ID-easy, UACP has only 0.0196 ms of feedback airtime, compared with 0.2048 ms and 0.8192 ms for the $N_g=16$ and $N_g=4$ sparse baselines, respectively, and 3.2768 ms for oracle. This corresponds to a 99.4% and 97.6% overhead reduction relative to oracle and Wi-Fi/D-Anchor with $N_g=4$. This incurs a negligible loss in prediction and system-level performance with NMSE of -31.6 dB and EVM of 3.75%. Moreover, the substantially lower feedback cost yields the highest end-to-end throughput at 165.07 Mbps, i.e., 83.4% above oracle and 17.4% above the most aggressive sparse baselines.

When the channel becomes more frequency-selective (ID-hard), UACP automatically moves to a denser feedback setting. Specifically, at only 0.1 ms of feedback airtime, UACP still reduces overhead by 96.9% with respect to oracle, yet it achieves 2.35×10^{-4} BER. Importantly, UACP outperforms Wi-Fi/D-Anchor with $N_g=16$, and improves upon Wi-Fi with $N_g=4$ despite using $8.2\times$ less feedback airtime. D-Anchor with $N_g=4$ remains slightly better on BER/EVM/NMSE, but only by incurring $8.2\times$ more feedback airtime. As for the throughput, UACP still offers the best operating point at 151.65 Mbps.

Once the channel departs from the training distribution (OOD), reliability is more important than compression. As such, UACP switches to full feedback, but it even slightly improves performance over oracle, achieving a BER of 2.25×10^{-4} , EVM 4.30%, and NMSE -39.0 dB, which is consistent with the denoising effect of the evidential predictor on noisy CFR observations. In contrast, Wi-Fi/D-Anchor with $N_g=16$ degrade to $8.40 \times 10^{-4}/7.20 \times 10^{-4}$ BER and 10.40%/9.35% EVM.

Table 1: Throughput (Mbps) across three regimes.

Method	ID-Easy	ID-Hard	OOD
Oracle	90.00	90.00	90.00
Wi-Fi/D-Anchor ($N_g=4$)	130.50	130.50	130.49
Wi-Fi/D-Anchor ($N_g=16$)	140.62	140.62	140.39
UACP	165.07	151.65	90.00

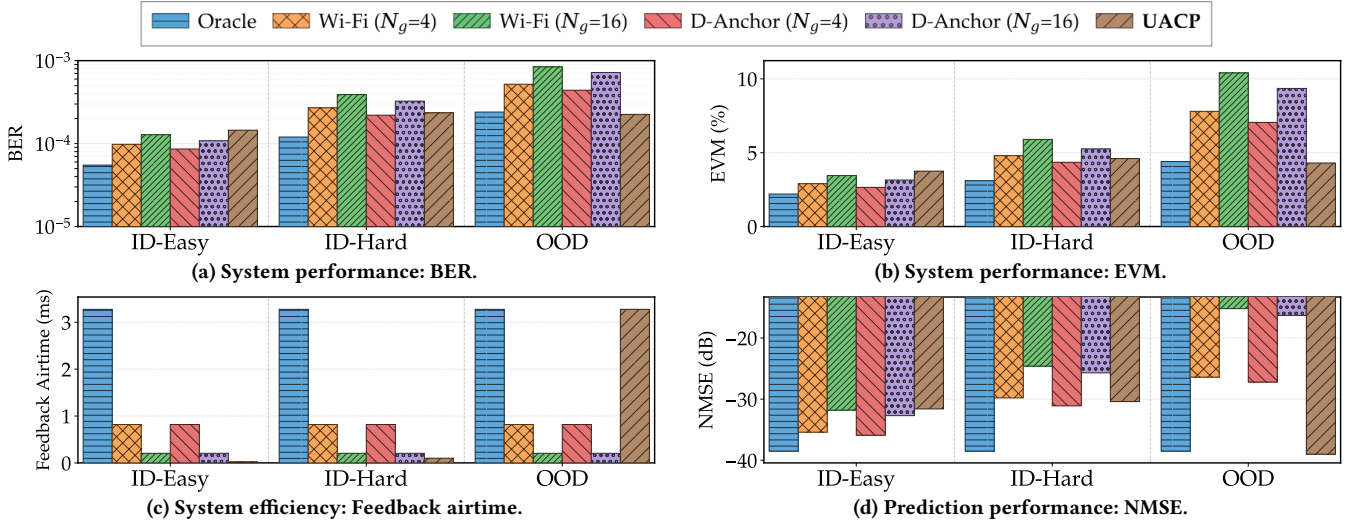


Figure 10: Trade-off between performance and efficiency. The first row demonstrates two wireless system-level performance metrics, while the second row shows two efficiency measures.

5.4 Adaptation to Regime Variations

A distinctive feature of UACP is that it closes the loop: uncertainty is not only reported, but immediately translated into a new feedback action through Equation (14). This makes UACP fully adaptable to changing wireless conditions. Figure 11 shows the time evolution of the downstream EVM, the subcarrier grouping factor selected by UACP, the reconstruction NMSE, and the system-level BER. In the first phase, where the channel is frequency-smooth, UACP converges to a high grouping factor $N_g=128$, keeping the feedback highly-sparse (only 1 subcarrier every 128 is reported). In the next window, the delay spread increases, making the channel highly frequency-selective. UACP recognizes this change and decreases the grouping factor to $N_g=16$ to guarantee accurate channel inpainting and maintain good communication performance, as shown by the NMSE and BER/EVM metrics. The following phases also follow the same trend. When the regime becomes OOD, the epistemic uncertainty exceeds the set threshold and UACP reacts by densifying the feedback to prevent performance drop. Overall, Figure 11 shows that the uncertainty produced by UACP is not only meaningful, but also actionable. Specifically, UACP can anticipate when sparse feedback becomes unsafe and requests additional subcarriers before the transmitter commits to poor precoding performance.

6 Related Work

Reducing MIMO channel sounding overhead has been an active research area in Wi-Fi and cellular systems. Under IEEE 802.11, each channel sounding round occurs at a fixed 10 ms [6] interval regardless of channel dynamics. The CFR is compressed before transmission using singular value decomposition (SVD) and Givens rotation. In cellular systems, codebooks are used for compression. Finding more effective compression mechanisms that improve system scalability when increasing the number of antennas and subcarriers has been a very active research area. Wen et al. [24] propose to compress the CSI using a convolutional encoder into a low-dimensional codeword, while [25] exploits the known sparsity of the massive MIMO channel in the angular-delay domain to

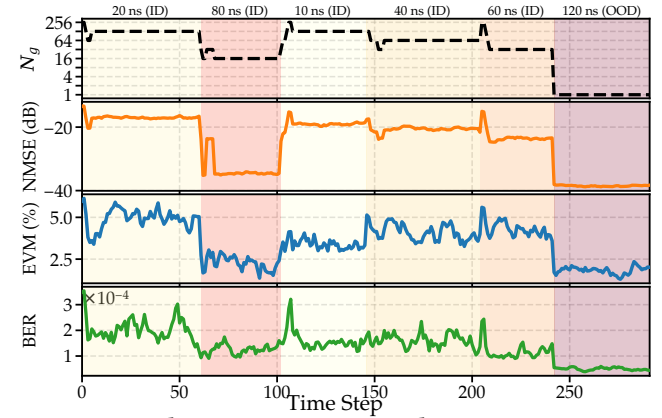


Figure 11: Adaptation to regime changes. Communication regimes are color-coded according to their degree of frequency-selectivity.

design a model-driven neural network encoder that achieves better CFR reconstruction accuracy than purely data-driven autoencoders at the same compression ratio. SplitBeam [2] reduces the receiver-side computation burden by designing a small compressor and offloading the heavier decoder to the transmitter side, while LB-SciFi [18] reduces feedback overhead using online reinforcement learning to adaptively select the compression type and feedback format according to channel conditions and network state.

Other complementary approaches in the literature propose to reduce sounding overhead by adapting the feedback rate to the channel conditions, either in the time domain by controlling how often sounding occurs, or in the frequency domain by controlling which subcarriers should be reported within each sounding round. For example, Bejarano et al. [3] track per-user channel through phase and magnitude variation history that allows the transmitter to predict a threshold for variations and suppress entire sounding rounds for users whose channel is static. This approach relies on coarse channel statistics that may not generalize well across diverse propagation environments. Ma et al. [11] analytically derives

optimal sounding balancing the airtime cost of sounding overhead and the CSI accuracy degradation due to channel aging. Rumman et al. [17] train a CNN at the receiver to predict the next CSI feedback from the historical observations. The receiver compares the prediction against the actual measurement and skips the sounding round entirely when cumulative prediction error remains below a threshold. Su et al. [19] developed a K-nearest neighbors model at the transmitter to estimate per-client throughput by correlating sequential channel measurements and finding the best match in a dataset of channel correlation and throughput pairs. Zhang et al. [26] use the medium access control (MAC) layer feedback overhead and the PHY layer channel capacity degradation to find the optimal sounding period. While effective, all approaches operate at a coarse temporal granularity – the entire sounding round is either performed or skipped – with no control over which specific subcarriers are reported within a given round. Notably, these time-domain approaches are orthogonal to the proposed framework: they decide *whether* to sound, while the proposed approach decides *what* to report within each sounding round. Therefore, they can be combined into a two-tier adaptive sounding strategy.

In the frequency domain, Wang et al. [22] uniformly subsample the CSI matrix and recover the remainder via a DNN, but with a fixed dropping pattern and never adapted to the dynamic channel variations. Endovitskiy et al. [4] addressed this limitation by proposing a dynamic anchor subcarrier selection algorithm. The receiver selects which subcarriers are fed back at runtime using a dynamic shortest-path algorithm on the current channel realization. Zhou et al. [27] instead merges adjacent subcarriers and extrapolates the full CSI at the transmitter, preserving all information at the cost of a fixed compression policy. Unlike these prior approaches, UACP uses uncertainty theory to drive adaptive subcarrier selection, explicitly distinguishing between aleatoric uncertainty caused by intrinsically noisy channels and epistemic uncertainty caused by channel conditions outside the model's training distribution.

7 Conclusions

In this paper, we presented UACP, an uncertainty-aware channel prediction and feedback adaptation framework for MIMO systems. By casting CFR prediction as evidential regression, UACP jointly predicts the channel and quantifies aleatoric and epistemic uncertainty, allowing the transmitter to distinguish hard channels from out-of-distribution ones and adapt feedback accordingly. UACP lowers feedback airtime by 92.7% and improves throughput by 21.3% over Wi-Fi ($N_g=4$), while paying only a 0.33% average EVM penalty. Under out-of-distribution shift, it falls back to full feedback and improves BER by 6.3% relative to an oracle precoder.

Future work includes coupling subcarrier-level adaptation with adaptive sounding intervals, so that the system can jointly decide when sounding should occur and what should be reported, as well as enabling online adaptation under persistent distribution shifts.

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